

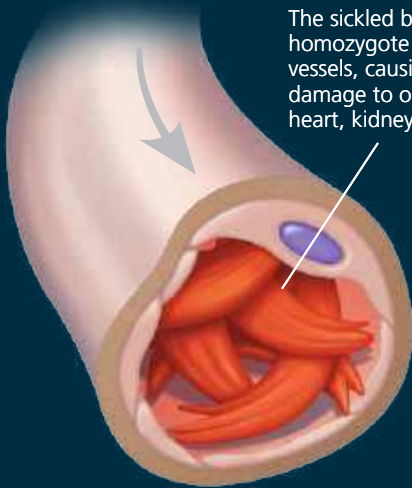


Infected mosquitoes spread malaria when they bite people. (See Figure 28.16.)

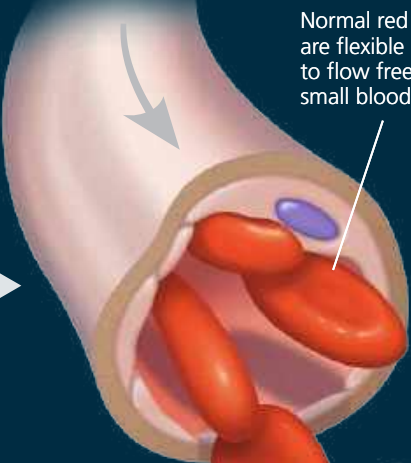
Effects on Individual Organisms

- The formation of sickled red blood cells causes homozygotes with two copies of the sickle-cell allele to have sickle-cell disease.
- Some sickling also occurs in heterozygotes, but not enough to cause the disease; they have sickle-cell trait. (See Figure 14.17.)

The sickled blood cells of a homozygote block small blood vessels, causing great pain and damage to organs such as the heart, kidney, and brain.

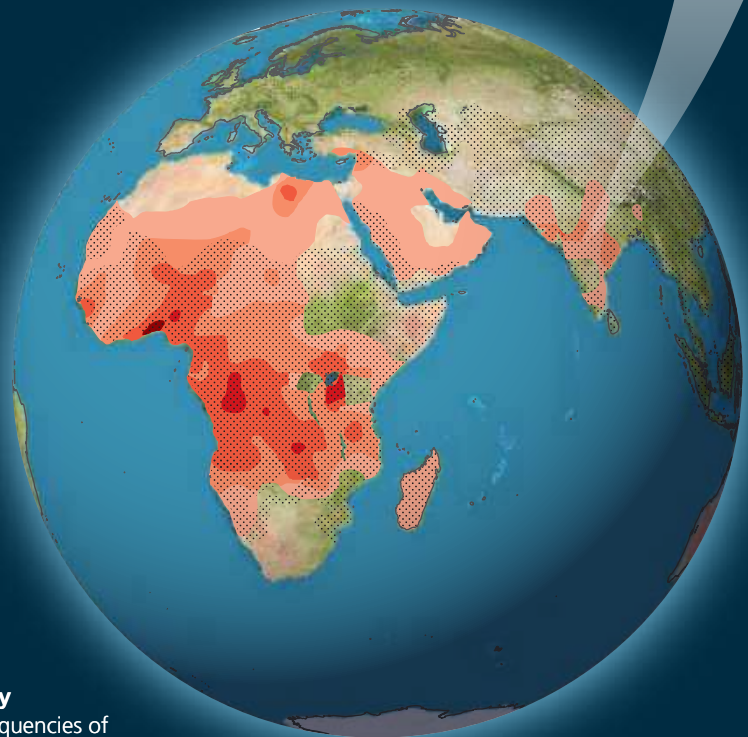


Normal red blood cells are flexible and are able to flow freely through small blood vessels.



Evolution in Populations

- Homozygotes with two sickle-cell alleles are strongly selected against because of mortality caused by sickle-cell disease. In contrast, heterozygotes experience few harmful effects from sickling yet are more likely to survive malaria than are homozygotes.
- In regions where malaria is common, the net effect of these opposing selective forces is heterozygote advantage. This has caused evolutionary change in populations—the products of which are the areas of relatively high frequencies of the sickle-cell allele shown in the map.



Key

Frequencies of the sickle-cell allele

- 3.0–6.0%
- 6.0–9.0%
- 9.0–12.0%
- 12.0–15.0%
- >15.0%

 Distribution of malaria caused by *Plasmodium falciparum* (a parasitic unicellular eukaryote carried by mosquitoes)

MAKE CONNECTIONS

In a region free of malaria, would individuals who are heterozygous for the sickle-cell allele be selected for or selected against? Explain.